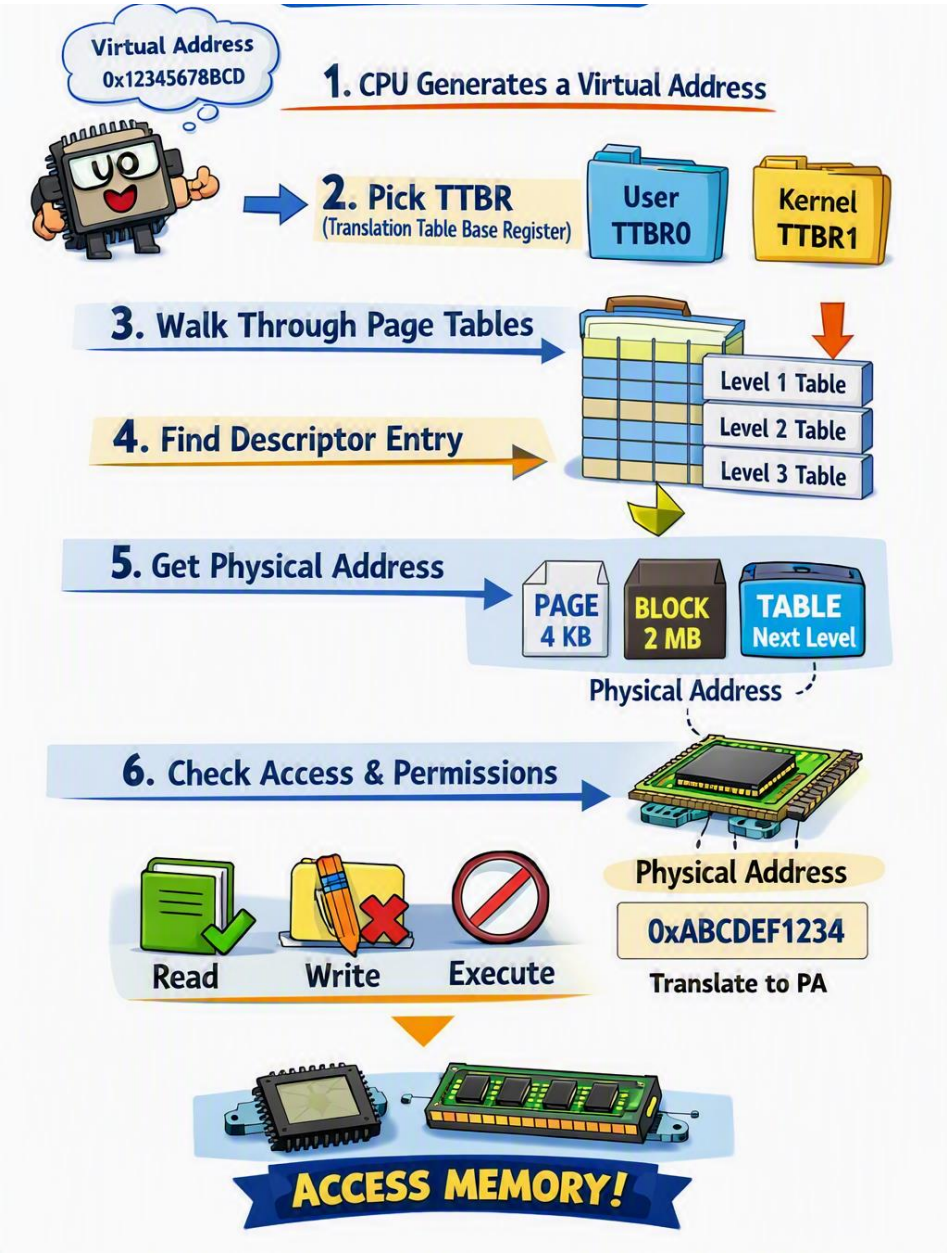
MMU Descriptor Bit-Level Summary

(ARMv8-A, 4KB Granule)



1. Block Descriptor (L1 / L2) — Bit-Wise Description

Descriptor Bits (Block: 1GB at L1, 2MB at L2)

[63]	Valid bit (must be 1)
[62:55]	Reserved
[54:52]	AttrIndx (MAIR attribute index)
[51:50]	Reserved
[49]	nT (reserved for table)
[48]	nG (not global)
[47:12]	Output address (physical base of block)
[11]	Contiguous hint
[10]	AF (Access Flag)
[9:8]	SH (Shareability: 00 NS, 10 Outer, 11 Inner)
[7:6]	AP (Access permissions)
[5]	NS (Non-secure)
[4:2]	Reserved
[1:0]	Descriptor type = 01 (Block)

2. Page Descriptor (L3) — Bit-Wise Description

Descriptor Bits (Page: 4KB)

[63]	Valid bit (must be 1)
[62:55]	Reserved
[54:52]	AttrIndx (MAIR attribute index)
[51:50]	Reserved
[49]	nT (reserved)
[48]	nG (not global)
[47:12]	Output address (physical base of 4KB page)
[11]	Contiguous hint
[10]	AF (Access Flag)
[9:8]	SH (Shareability)
[7:6]	AP (Access permissions)

[5]	NS (Non-secure)
[4:2]	Reserved
[1:0]	Descriptor type = 11 (Page)

3. Table Descriptor (L0 / L1 / L2) — Bit-Wise Description

Descriptor Bits (Table: points to next-level table)

[63]	Valid bit (must be 1)
[62:55]	Reserved
[54:12]	Next-level table address (physical base)
[11:2]	Next-level table address (continued)
[1:0]	Descriptor type = 11 (Table)

Notes:
- No AttrIndx
- No AP
- No SH
- No AF
- No UXN/PXN
- No permissions or memory attributes

4. Virtual Address Bit-Wise Description (48-bit VA, 4KB Granule)

VA[47:39]	Level 0 index (9 bits)
VA[38:30]	Level 1 index (9 bits)
VA[29:21]	Level 2 index (9 bits)
VA[20:12]	Level 3 index (9 bits)
VA[11:0]	Page offset (4KB page)

Properties:

- Each index = 9 bits → 512 entries per table
- VA contains only indices + offset
- Descriptor bits are NOT part of the VA